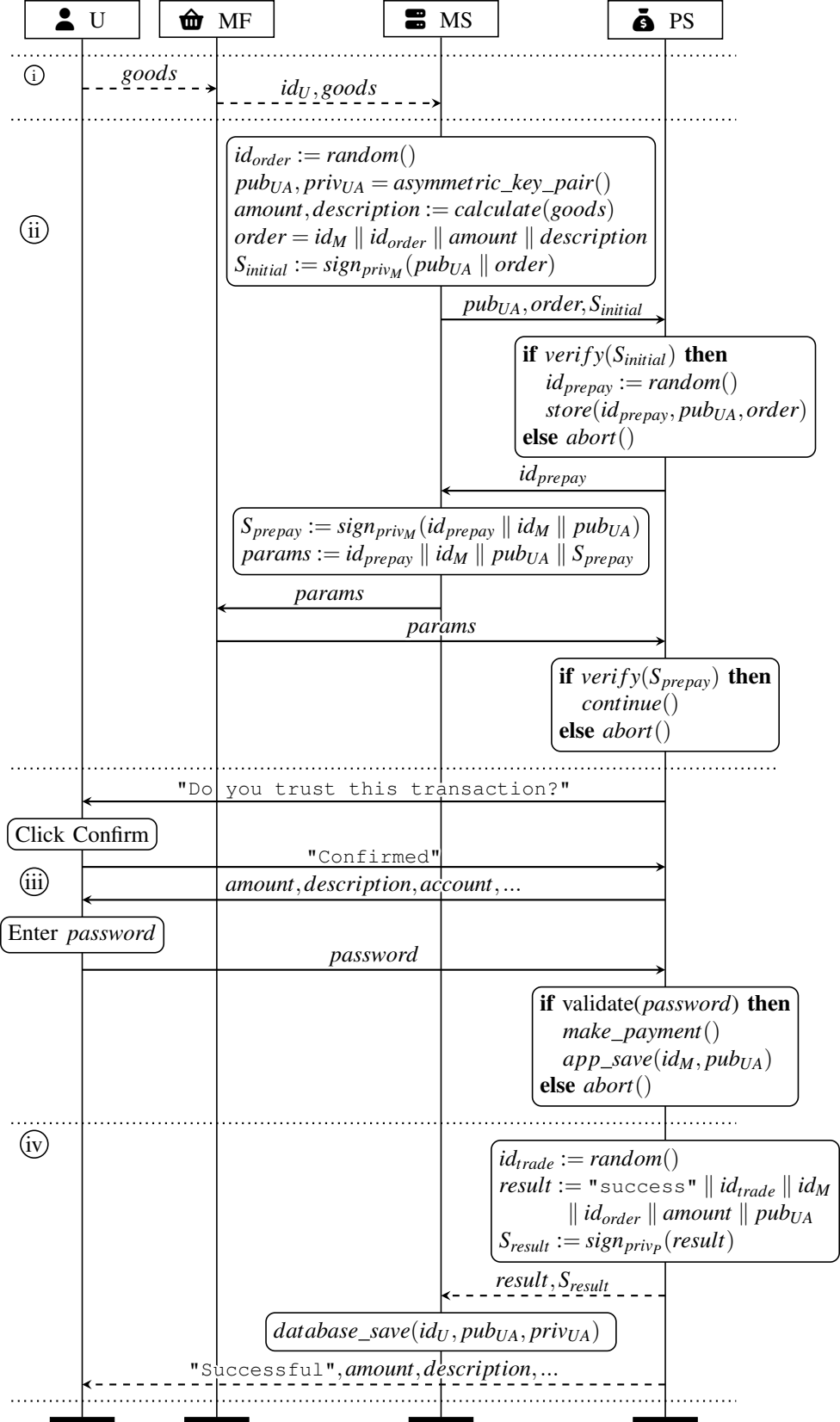

U
MF
MS
PS
i
goods
$id_U, goods$
ii
$id_{order} := random()$
$pub_{UA}, priv_{UA} = asymmetric_key_pair()$
$amount, description := calculate(goods)$
$order = id_M \parallel id_{order} \parallel amount \parallel description$
$S_{initial} := sign_{priv_M}(pub_{UA} \parallel order)$
$pub_{UA}, order, S_{initial}$
if $verify(S_{initial})$ **then**
$id_{prepay} := random()$
$store(id_{prepay}, pub_{UA}, order)$
else $abort()$
id_{prepay}
$S_{prepay} := sign_{priv_M}(id_{prepay} \parallel id_M \parallel pub_{UA})$
$params := id_{prepay} \parallel id_M \parallel pub_{UA} \parallel S_{prepay}$
$params$
$params$
if $verify(S_{prepay})$ **then**
$continue()$
else $abort()$
"Do you trust this transaction?"
Click Confirm
"Confirmed"
iii
$amount, description, account, ...$
Enter *password*
$password$
if validate($password$) **then**
$make_payment()$
$app_save(id_M, pub_{UA})$
else $abort()$
iv
$id_{trade} := random()$
$result :=$ "success" $\parallel id_{trade} \parallel id_M \parallel id_{order} \parallel amount \parallel pub_{UA}$
$S_{result} := sign_{priv_P}(result)$
$result, S_{result}$
$database_save(id_U, pub_{UA}, priv_{UA})$
"Successful", $amount, description, ...$